

Qatar-2b

R-1164

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_160408 · created 2026-07-31T02:44:07+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457486.87013 +/- 0.00026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.227 +/- 0.066
Transit depth (Rp/Rs)^2	5.17 +/- 2.99 [%]
Orbital Inclination (inc)	88.09 +/- 1.74
Airmass coefficient 1 (a1)	2.79 +/- 1.59
Airmass coefficient 2 (a2)	-0.63 +/- 0.29
Scatter in the residuals of the lightcurve fit is	52.25 %
Best Comparison Star	None
Optimal Aperture	4.1
Optimal Annulus	7.86
Transit Duration (day)	0.0613 +/- 0.0027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.227 $\pm$ 0.066 vs 0.16327 (deviation 0.97 σ)
Duration fit vs published	0.0613 d vs 0.0767 d (deviation 5.69 σ)
Mid-time O-C	-0.5 min
Depth SNR	3.4
Residual scatter	52.25 %

Light curve

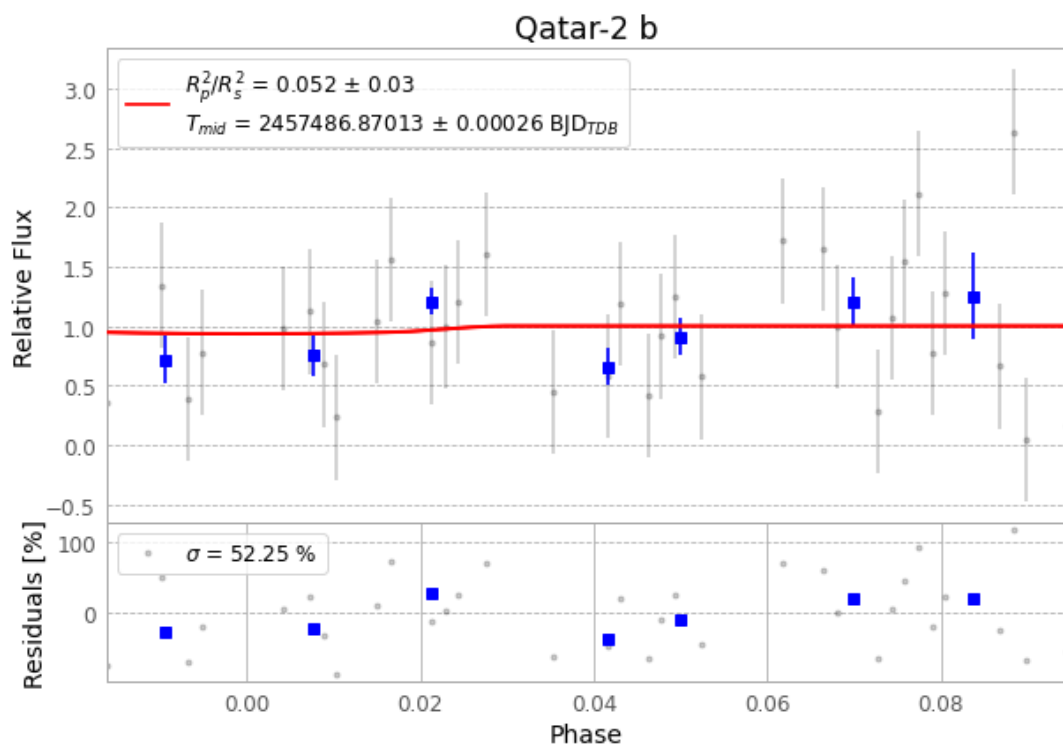


Figure 1 — FinalLightCurve_Qatar-2 b_2016-04-08.png (SHA-256 3f9f849516213be9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [359, 407], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 921b66ec1e9372d0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), Qatar-2160408080612.FITS → Qatar-2160408115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-160408.log (8c001f283fe3...), FinalLightCurve_Qatar-2 b_2016-04-08.png (3f9f84951621...), FinalLightCurve_Qatar-2 b_2016-04-08.csv (82501952040e...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 02:44Z.